A Comparative Study of Deep Learning Approaches for Aviation Safety Report Classification

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1 Introduction

Globally, Civil aviation authorities manually classify aviation safety reports according to the International Civil Aviation Organization (ICAO) Accident/Incident Reporting (ADREP) taxonomy. Manual processing leads to inconsistent labeling, processing delays, and missed opportunities for trend detection among other challenges. Among the studies that have approached this problem, a CMU-Africa Capstone project [5] demonstrated effective automation using a hybrid transformer-multi-LLM pipeline. Their approach succeeds in making a case for the use of transformers and LLMs for the task, successfully classifying most reports, but still fails to classify 26.94% of the aviation safety reports which were conservatively classified as **"OTHER"**. These failed classifications typically correspond to narratives with implicit, multi-factor, or weakly expressed causal structure where flat text classification lacks sufficient relational grounding.

We focus our analysis on these failed classifications to validate our approach. Assuming if we are able to classify these difficult to classify accident reports, we can confidently apply our approach to the entire dataset confident we'll achieve better classification across all classes..

2 Related Work

Aviation safety NLP has evolved through three main phases: classical text analytics, supervised/deep-learning classification, and domain-adapted transformer systems.

Early work established that incident narratives can be mined for operational safety insight using structured NLP and data-mining pipelines. These studies showed that narrative text supports meaningful categorization and trend analysis, while also exposing limits in semantic depth and interpretability in traditional approaches [14, 12].

More recent applied studies shifted toward supervised NLP and deep learning for report classification and injury-severity prediction. Across these studies, performance improved in practical settings, but outcomes remained sensitive to annotation quality, class imbalance, and dataset-specific reporting style [11, 8, 9].

In parallel, domain-adapted language models were proposed to better capture aviation terminology and context. Aviation-BERT and SafeAeroBERT provide evidence that aviation-specific pretraining improves representation quality for downstream safety tasks, while long-context transformer designs such as Longformer motivate architectures better suited to lengthy causal narratives [4, 1, 3].

Building on these strands, the CMU-Africa hybrid pipeline combines supervised Longformer prediction, semantic matching, and multi-LLM consensus under taxonomy constraints derived from ICAO ADREP. This system substantially improves coverage but still leaves a meaningful subset of reports in the **OTHER** class, motivating targeted reclassification strategies such as ours [5, 7, 2].

Overall, prior work supports our direction: use domain-aware language modeling and structured decision logic to improve fine-grained aviation occurrence classification under the ADREP framework.

3 Methods

The dataset used throughout this work is the ASN Dataset, consisting of 19,675 aviation incident/accident reports scraped from the Aviation Safety Network[2]. Each record contains a free-text narrative field (the primary evidence for classification) along with metadata fields. The target label space contains 37 ECCAIRS occurrence categories[7], though the pre-existing automated system only confidently assigns labels to a small subset of them, leaving the vast majority of reports unclassified

We explore 3 ways to improve upon the previous work, a fine-tuned end to end RoBERTa architecture, a four-layer causal refinement module and a [**INSERT APPROACH 3 DESCRIPTION**]

3.1 Approach 1; A RoBERTa Transformer Model

3.1.1 Architecture Evolution

The system architecture used in the CMU-Africa study consisted of a multi-stage pipeline where a RoBERTa-based Question Answering model extracted key text spans, followed by a retrieval-based classification step utilizing all-MiniLM-L6-v2 cosine similarity against ECCAIRS taxonomy embeddings. This approach suffered from a severe information bottleneck, discarding approximately 90% of the raw narrative context and yielding a 71.6% "Unknown" (UNK) classification rate.

To resolve this limitation, the architecture was redesigned as an end-to-end Sequence Classification pipeline. The new framework directly feeds up to 512 tokens of the raw aviation incident narrative into a roberta-base Transformer model. A classification head with 37 output nodes (corresponding to the ECCAIRS occurrence categories) maps the full narrative directly to the target classification. To mitigate the severe class imbalance present in the domain dataset, the network is optimized using Cross-Entropy Loss with balanced class weights.

3.1.2 Experiments

To incrementally evaluate and optimize the architecture, a fixed, stratified test set of 2,000 high-confidence labeled rows was established prior to training. The following core experiments have been conducted to date:

Experiment 1 (Baseline Audit): Evaluated the legacy cosine-similarity retrieval system. Results confirmed poor coverage, establishing a 71.6% UNK(Unkown) rate baseline on the full corpus and highlighting the critical need for a supervised deep learning approach.

Experiment 2 (End-to-End RoBERTa): Fine-tuned the roberta-base architecture on raw narratives utilizing high-confidence rule-based labels. This fundamental architectural shift successfully reduced the corpus UNK rate to 0.00% while achieving a near-perfect Macro F1 score of 0.9979 and Top-5 accuracy of 1.000 on the test set.

Experiment 3 (Domain-Specific Pre-training): Swapped the general roberta-base backbone for NASA-AIML/MIKA_SafeAeroBERT[10] to test the hypothesis that aviation-domain priors would improve understanding of confusable categories. Counter-intuitively, this degraded performance (Macro F1 dropped to 0.9027) due to catastrophic performance drops on extreme minority classes (e.g., Abnormal Runway Contact), leading to the rejection of the domain-specific model.

Experiment 4 (Cross-Encoder Reranking): Transformed the multi-class formulation into a binary sequence pair task by jointly encoding the narrative and category description. This approach suffered from extreme $O(N)$ inference latency (5 seconds per report) and a severe Macro F1 collapse (0.6893) due to inadequate hard-negative mining in the highly imbalanced vector space.

Experiment 5 (Iterative Self-Training): Attempted dataset augmentation via pseudo-labeling by querying the unlabeled corpus with the Experiment 2 model and injecting 13,191 high-confidence (≥ 0.80) predictions back into the training pool. While mean model confidence increased overall, Macro F1 decreased to 0.9180 as the overwhelming volume of dominant-class pseudo-labels introduced signal noise that impaired the classification of algorithmic edge cases. Ultimately, the rigorous ablation and experimental sequence confirmed that the Experiment 2 standard roberta-base architecture remains the most robust, accurate, and efficient state-of-the-art solution for this taxonomy.

3.1.3 Future Work & Upcoming Experiments

While the single-class prediction performance of the Experiment 2 framework is exceptionally high, the real-world nature of aviation incidents frequently involves compounding concurrent failures (e.g., severe icing leading to a loss of control). The remaining architectural validation planned for this pipeline is:

Experiment 6 (Multi-Label Classification): The network's output head will be transitioned from CrossEntropyLoss with a Softmax activation to BCEWithLogitsLoss utilizing a Sigmoid activation. This will enable the model to output a continuous multi-hot vector of length 37, capturing multiple concurrent event types per narrative. The primary evaluation metric for this final phase will shift to mean Average Precision (mAP) to dynamically reflect multi-label predictive capabilities.

3.2 Approach 2; Causal Refinement

In this approach we use a four-layer causal refinement module that acts exclusively on OTHER-labelled reports. The goal of Layer 1 is to extract a structured event tuple of the form

Event = (ACTOR, SYSTEM, PHASE, TRIGGER, OUTCOME)

from each raw aviation narrative. These structured tuples are the input to Layer 2 (causal graph construction), Layer 3 (graph reasoning), and Layer 4 (ADREP reclassification). Without a reliable extraction layer, downstream components cannot function making this the foundational piece of the pipeline. The extraction problem is a Named Entity Recognition (NER) task framed as token-level sequence labeling, where each sub-word token of a narrative is assigned a BIO (Begin–Inside–Outside) label. PHASE and OUTCOME are obtained directly from existing structured columns in the dataset ($\geq 98\%$ and 100% coverage respectively), so the NER model focuses on the three text-only entities: ACTOR, SYSTEM, and TRIGGER.

3.2.1 Silver-Label Generation

Since the dataset contains no gold-annotated NER labels, training data is generated via a silver-labeling strategy. Three sets of domain-curated regex patterns are applied to each narrative to extract character-level entity spans:

ACTOR patterns — cover specific crew roles (Captain, First Officer, PIC, SIC, ATC, Flight Crew), ATC roles, and first-person pilot narrations ("I noticed...", "we decided...")

SYSTEM patterns — cover navigation systems (GPS, ILS, RNAV, FMS), safety systems (TCAS, GPWS, TAWS), autopilot/automation, communications (radio, transponder), and aircraft components (engine, landing gear, flaps)

TRIGGER patterns — cover causal language ("due to", "failed to", "resulted from"), human factors (fatigue, distraction, situational awareness loss), environmental triggers (turbulence, icing, IMC), and mechanical failures (engine failure, malfunction)

Overlapping spans are resolved by keeping the longest match. The resulting character-level spans are then aligned to BERT sub-word token positions to produce BIO-tagged sequences, forming the training set for fine-tuning. This silver-labeling strategy is deliberate: the BERT model is expected to generalise beyond the rules by learning contextual cues from the BERT pre-training that pattern matching alone cannot capture.

3.2.2 Model Architecture

The model follows the standard Transformer + Token Classification Head architecture:

$$[\text{narrative tokens}] \rightarrow \text{BERT Encoder} \rightarrow H \in \mathbb{R}^{L \times 768} \rightarrow \text{Linear } (768 \rightarrow 7) \rightarrow \text{BIO logits}$$

The classification head maps each token's hidden state to 7 labels: O, B-ACTOR, I-ACTOR, B-SYSTEM, I-SYSTEM, B-TRIGGER, I-TRIGGER.

Two pre-trained encoders are evaluated:

- **BERT-base-uncased**. A general-purpose BERT[6]

- **SafeAeroBERT** (NASA-AIML/MIKA_SafeAeroBERT). A domain-specific BERT pre-trained on ASRS and NTSB aviation safety narratives; the same type of data we work with.[1]

Both models share 108.9M trainable parameters and are trained under identical hyperparameters for a fair comparison.

In the training, class weighting is applied to the cross-entropy loss to mitigate the heavy imbalance between the O label (which accounts for the vast majority of tokens) and entity labels. Weights are normalised so that O receives a weight of 1.0, and entity tokens receive up to 20×. Additionally long narrative handling: narratives exceeding 256 tokens are split into overlapping windows (stride = 128 token-equivalents) so entities near chunk boundaries are not lost.

3.2.3 Experiments

The rule-based silver-label extractor itself serves as the baseline. It is deterministic and interpretable, but it is limited to patterns explicitly enumerated in the pattern lists — it cannot generalise to unseen phrasing or contextual variations. The BERT models are expected to improve upon it by learning latent contextual signals from pre-training.

The following experiments are planned for the remainder of the project:

- CRF decoding layer — add a Conditional Random Field (CRF) on top of the BERT hidden states to enforce valid BIO transition constraints (e.g., no I-ACTOR without a preceding B-ACTOR), which the current linear head does not guarantee.
- Coreference resolution for ACTOR recall — the current model misses actors expressed through first-person pronouns in non-pattern-matched contexts. Integrating a lightweight coreference resolution step (e.g., coreferee or neuralcoref) to resolve "we/I" → pilot role before extraction could improve ACTOR recall.
- LLM-augmented silver labels — replace or supplement the rule-based silver labels with structured extractions from an LLM (e.g., Gemini) using chain-of-thought prompting, then fine-tune BERT on the richer labels.
- Active learning — use model uncertainty (token-level entropy) to select the most informative reports for manual annotation, enabling targeted quality improvement with minimal annotation effort.
- Layer 2 integration — pass the extracted event tuples from extracted_events.json into a causal DAG construction pipeline using torch_geometric, with typed nodes (ACTOR, SYSTEM, PHASE, TRIGGER, OUTCOME) and directed causal edges.

3.3 Approach 3

To compare the performance of Approach 2, a domain classification (ontology) combined with a hybrid annotation strategy was used. This approach was necessary because the corpus contains highly domain-specific terminology that generic NER models or simple regex patterns do not capture. A domain ontology and a hybrid annotation pipeline was developed that integrated manual annotation with LLM-guided labeling, followed by human evaluation on a sample of 2,500 corpus texts.

To improve named entity recognition (NER) in aviation narratives, a manual curation of large domain lexicons was done for each entity type. The lexicon was built from the combination of terminology in the Aviation Safety Network (ASN) data repository and with synonyms used in the domain. The resulting vocabulary contains hundreds of domain specific entity mentions, that was considered to help significantly improve recall over standard NER models trained on general corpora.

3.3.1 Hybrid Annotation Pipeline

Because the ASN dataset contains no gold-standard NER labels, we constructed a hybrid annotation dataset of about 2,000 labeled narratives using a combination of two strategies. A preliminary experiment was conducted on a small dataset of about 20 manually inspected narratives to evaluate and help choose candidate BERT transformer NER models (models considered were `bert-base-uncased`, `roberta-base`, `distilbert-base-uncased`, `NASA-AIML/MIKA_SafeAeroBERT`) and assess

their ability to recognize aviation-specific entities. To establish a reliable supervision signal, 1,100 narratives were manually annotated using the BIO tagging scheme (B-ACTOR/I-ACTOR, B-SYSTEM/I-SYSTEM, B-TRIGGER/I-TRIGGER), ensuring high-quality labels for complex aviation phrases not captured by simple patterns.

To scale the dataset beyond manual labeling that takes time given the scope of the project, we employed LLM-assisted annotation with human validation, prompting a large language model with structured extraction instructions to identify candidate ACTOR, SYSTEM, and TRIGGER spans, this later required validation by manual checking. The LLM output was post-processed using domain lexicons and token alignment to produce BIO labels.

Regex patterns and domain lexicons were used to automatically label further entity mentions, providing silver labels for narratives not manually annotated. Through this hybrid process, we generated approximately 2,000 labeled narratives, consisting of 1,000 manually annotated examples and 1,000 LLM-assisted and rule-verified examples.

3.3.2 Aviation-Specific BIO Label

Unlike standard NER datasets (for instance, the ones that use the following labels: PER, LOC, ORG, etc.) [13], aviation safety narratives contain domain-specific entities that describe operational events. Therefore, we define a custom label class aligned with the causal event structure in mind, as shown below in table 1.

Label	**Meaning**
O	Outside any entity
B-ACTOR / I-ACTOR	Human agent involved in the event
B-SYSTEM / I-SYSTEM	Aircraft component or system
B-TRIGGER / I-TRIGGER	Immediate causal trigger
B-PHASE / I-PHASE	Flight phase
B-OUTCOME / I-OUTCOME	Final event outcome

Table 1: Entity labels and their meanings in the annotation scheme.

3.3.3 Experiment

To evaluate the feasibility of extracting entities from aviation safety narratives, we first performed a small-scale training on approximately 20 narratives to compare candidate BERT NER models and verify their ability to capture aviation-specific terminology. Based on this evaluation, we adopted a token classification framework using the BIO tagging scheme to label entities. Because the dataset does not contain gold NER annotations, we constructed a hybrid labeled dataset of roughly 2,000 narratives from scraped data provided. Of these, about 1,000 reports were manually annotated to ensure high-quality supervision for complex aviation phrases, while the remaining examples were generated using a combination of domain lexicon matching and LLM-guided annotation with validation done by a human. The resulting BIO-tagged sequences were used to fine-tune transformer encoders for token classification, with class-weighted cross-entropy loss applied to address the strong imbalance between entity tokens and the dominant `O` class.

4 Results

This section summarizes quantitative results for each approach and compares them against the corresponding baselines.

4.1 Approach 1 (RoBERTa Transformer Model) Results

Table 2: Comparative Results of Aviation Incident Classification Architectures

Experiment	Architecture / Approach	UNK Rate ↓	Macro F1 ↑	Weighted F1 ↑	Mean Conf. ↑	Top-5 Acc. ↑
EXP-1	Cosine Similarity (Baseline)	71.60%	N/A*	N/A*	**0.9993**	N/A*
EXP-2	End-to-End RoBERTa	**0.00%**	**0.9979**	**0.9975**	0.9945	**1.0000**
EXP-3	AeroBERT Domain Model	**0.00%**	0.9027	0.9947	0.9897	0.9995
EXP-4	Cross-Encoder Reranking	**0.00%**	0.6893	0.8362	0.8634	0.9945
EXP-5	Iterative Self-Training	**0.00%**	0.9180	0.9952	0.9985	**1.0000**

*N/A for baseline as the test set was constructed from rule-based baseline high-confidence predictions.

4.2 Approach 2 (NER) Results

Both models were evaluated on a held-out test set (15% of 173 reports). Weighted F1, weighted precision, and weighted recall are computed using seqeval, the standard library for sequence labeling evaluation.

Table 3: NER performance comparison on the held-out test set.

Metric	BERT-base	SafeAeroBERT	Δ (Aero – BERT)
Test F1 (weighted)	0.741	0.830	+0.089
Precision	0.655	0.786	+0.131
Recall	0.857	0.881	+0.024
Test Loss	0.690	0.495	-0.195
Best Val F1	0.731 (epoch 9)	0.758 (epoch 10)	+0.027

SafeAeroBERT is the clear winner, outperforming BERT-base by +8.9 F1 points on the test set. The improvement is most pronounced in precision (+13.1 points), suggesting the domain-specific pre-training allows the model to be more selective by gaining the ability to identify true entity spans with fewer false positives.

Figure 1: Figure 2: BERT-base vs. SafeAeroBERT on F1 score, precision, and recall on the held-out test set

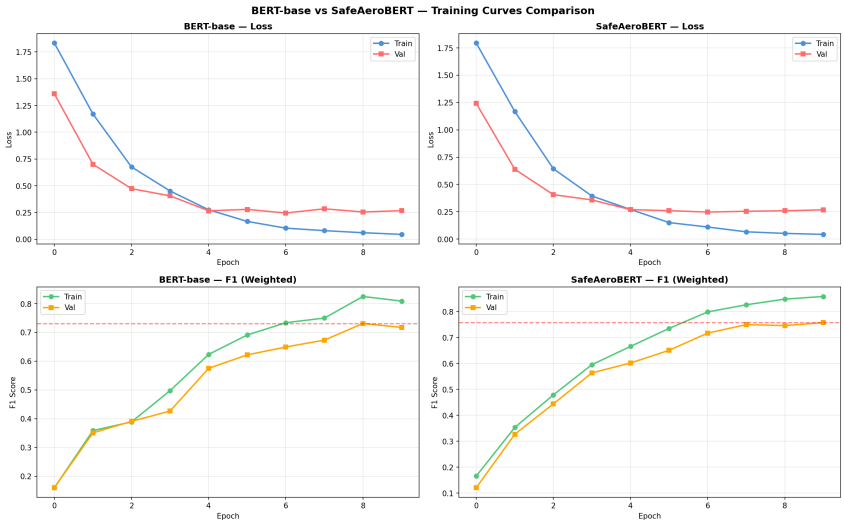
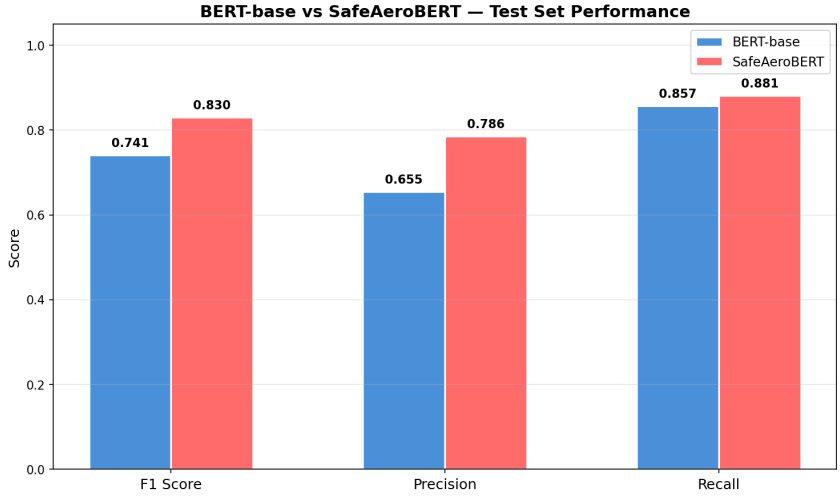


Figure 2: Training and validation loss (top) and F1 (bottom) for BERT-base (left) and SafeAeroBERT (right) over 10 epochs.



4.3 Approach 3 (NER with Human Curated Data) Results

Table 4: Comparison of transformer models for aviation NER using BIO tagging and human-curated annotations.

Model	Eval Loss	Precision	Recall	F1-score	Epochs
bert-base-uncased	0.024	0.955	0.960	0.956	5
roberta-base	0.082	0.893	0.831	0.860	5
allenai/scibert_scivocab_uncased	0.060	0.879	0.880	0.879	5
distilbert-base-uncased	0.035	0.946	0.925	0.936	5
NASA-AIML/MIKA_SafeAeroBERT	0.021	0.957	0.971	0.964	5

Table 4 exhibits the performance of several BERT-based token classification models trained on the hybrid annotated dataset using BIO tagging. The results indicate consistently strong performance

across architectures, with the domain-specialized SafeAeroBERT achieving the highest overall F1-score and recall. Compared with the earlier silver-label experiments in Approach 2, these results demonstrate that human-curated annotations substantially improve extraction accuracy.

Figure 3 visualizes the precision and recall trade-offs across the fine-tuned and evaluated models. SafeAeroBERT achieves the highest recall while maintaining strong precision, indicating effective recognition of aviation entities without excessive false positives, this is expected as it is trained on aviation specific language[1]. DistilBERT also performs comparatively well while offering faster inference efficiency. These results suggest that domain-adapted (fine-tuned) encoders combined with human curated annotations significantly improve entity extraction quality.

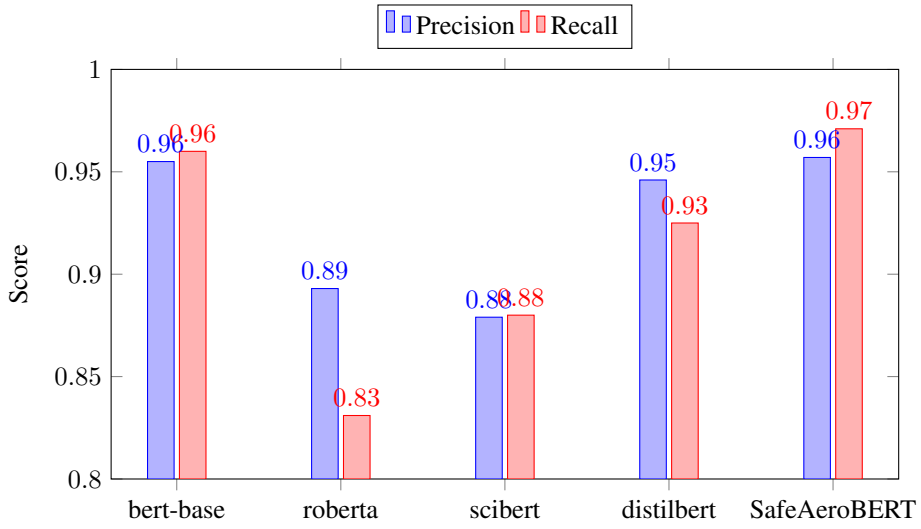


Figure 3: Grouped bar chart comparing precision and recall across evaluated NER models.

5 Ablation Study (if applicable)

You may have several ideas and components of your project pipeline you wish to explore; Since some of these components may be more or less important, you should run isolated experiments with them and find how each part is affecting your final result.

6 Discussion and Conclusion

From the **RoBERTa Transformer Model(Approach 1)**, we demonstrate that while mitigating the severe "Unknown" (UNK) classification rate in aviation incident reports is achievable through deep learning, the most parsimonious approach yielded the most robust results. The transition from a lossy extraction pipeline to an end-to-end roberta-base sequence classifier (EXP-2) successfully reduced the 71.6% UNK bottleneck to 0.00% without sacrificing accuracy. Critically, advanced interventions consistently underperformed this baseline: domain-specific pre-training (EXP-3) collapsed on rare classes, cross-encoder reranking (EXP-4) introduced prohibitive computational latency ($O(N)$ slowdown) alongside degraded accuracy, and pseudo-label augmentation (EXP-5) amplified signal noise from dominant classes, harming the model's ability to discern edge cases. These findings conclude that a standard, heavily-parameterized language model—when fine-tuned directly on raw narratives with balanced class weights—provides the optimal, state-of-the-art solution for this taxonomy, prioritizing inference speed and graceful generalization without the overhead of bespoke architectures.

We find when we use the **Causal Refinement(Approach 2)** that fine-tuning a domain-adapted BERT model on silver-labeled aviation narratives produces a reliable event extractor for the ACTOR, SYSTEM, and TRIGGER entities needed by the causal pipeline. SafeAeroBERT achieves a weighted F1 of 0.830 on the test set and 0.800 on manual annotations, with perfect precision — meaning it

does not invent entities that are not there. The hybrid extraction design proved to be a pragmatic and effective choice. Rather than attempting to learn PHASE and OUTCOME from text (where structured labels already exist), the model focuses its capacity on entities that genuinely require reading comprehension. This keeps the task tractable on a small dataset of 173 reports.

6.1 Limitations

- **Silver-label noise**: The 173 reports used here are a sample drawn from the full 68,000+ report corpus. While scaling to more reports is straightforward, the quality ceiling is set by the silver-labeling rules — if a pattern systematically mislabels a span, the model learns to replicate that error.
- **ACTOR recall**: First-person pilot narrations ("I noticed the engine was surging...") are partially missed because the rule patterns do not cover all pronoun-role associations. This is the main source of false negatives.
- **Silver-label ceiling**: Both models are ultimately trained on imperfect supervision. If the rule-based extractor systematically misses a certain entity type in a certain phrasing, the BERT model inherits that blind spot.
- **No causal relation extraction yet**: The current output is a flat entity tuple, not a causal graph. Edge types between entities (e.g., ACTOR caused TRIGGER led_to OUTCOME) are not yet modelled; this is the responsibility of Layer 2.

6.2 Challenges

- **BIO alignment with sub-word tokenization**: BERT tokenizes words into sub-word pieces (e.g., "TCAS" $\rightarrow$ ["TC", "##AS"]). Aligning character-level regex spans to sub-word token positions required careful offset mapping to correctly assign B- vs I- tags and ensure [CLS]/[SEP]/[PAD] tokens are masked from the loss with -100.
- **Class imbalance**: The O label accounts for the overwhelming majority of tokens in any narrative (>95%). Without explicit class weighting, the model converges to predicting O everywhere. Inverse-frequency weighting (capped at 20×) was critical to obtaining meaningful entity predictions.
- **Long narratives**: Some reports exceed 256 tokens. The overlapping window strategy (stride = 128 token-equivalents) ensures entities near chunk boundaries are not missed, but requires careful span re-indexing within each chunk.
- **Evaluation with seqeval**: Seqeval computes entity-level metrics (a span is only correct if both boundary tokens and the entity type match exactly), which is stricter than token-level accuracy. This is the appropriate metric for NER but takes care to interpret correctly alongside silver-label training.

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